Analyzing Factors Contributing to Severity of Vehicle Accidents

Chenchen Ma, Chenyun Ni, Koroby Wang, Ning Zhu

05/28/2024

I. Introduction

In each year, over 20 million vehicle accidents occur globally. In the United States, the reported amount is over 2.5 million per year, and the damage resulting from vehicle accidents is nearly 250 billion dollars (Karlin, 2024). The total yearly deaths in vehicle accidents in the United States reached 46,027 people in 2022 (Historical car crash deaths and rates, 2024). Clearly, vehicle accidents cause huge financial pressure and life risk for the United States. However, as the most common transportation method in the United States, driving is necessary in our daily lives. It is important to search for factors that affect the seriousness of injuries due to serious and fatal injuries that damage safety and finances more than minor injuries. In addition, vehicle accidents are caused by several contributing factors. Hence we need to search for several factors and consider these factors together. From several factors, speeding, vehicle age, and driver's fault attract our attention.

To reduce the accident cases and minimize the effects, it is necessary to analyze the vehicle accident data to figure out the relative elements. In this study, we will use crash-reported data to analyze and create a model to indicate the influence of speeding vehicle age, and driver's fault to the risk of fatal and serious injury, which might be helpful to future estimation.

II. Data description

For the dataset, we find Crash Reporting - Drivers Data which was published on data.gov, an official website of the GSA's Technology Transformation Services. This dataset reports details of all traffic accidents occurring on the roadways within Montgomery County. The reason that we picked the data about Montgomery County rather than the data about all the states is that the United States law-building differs in each county, and this dataset is well-built for analysis.

In this dataset, there are a total of 43 variables. We chose injury severity (Injury. Severity) as our response variable, in this variable, there are five states: no apparent injury, possible injury, suspected minor injury, suspected serious injury, and fatal injury. Since we are focusing on researching the individual injury, we removed the "no apparent injury", we also removed the "possible injury" as its definition is ambiguous: in crash data, the possible injury typically refers to situations where an injury is suspected but not visibly confirmed or diagnosed at the scene. In the remaining states, we set the response variable to binary variable: suspected minor injury as 0, suspected serious injury and fatal injury as 1. From the rest variables, as some of them have too many classes, such as weather, some of them are difficult to analyze, such as latitude, hence we choose two numerical variables: Speed. Limit, Vehicle. Year; and one binary variable: Driver. At.Fault. As the year of accident occurs is different between each data, we use the used year which is an accident occurring year minus vehicle produced year, instead of the variable Vehicle. Year.

III. Exploratory data analysis

In this section, we are going to analyze and investigate data sets and summarize their main characteristics. Since we have already chosen the three major variables, we are going to

figure out the relationship between each variable and our binary Y variable by visualizing the data first.

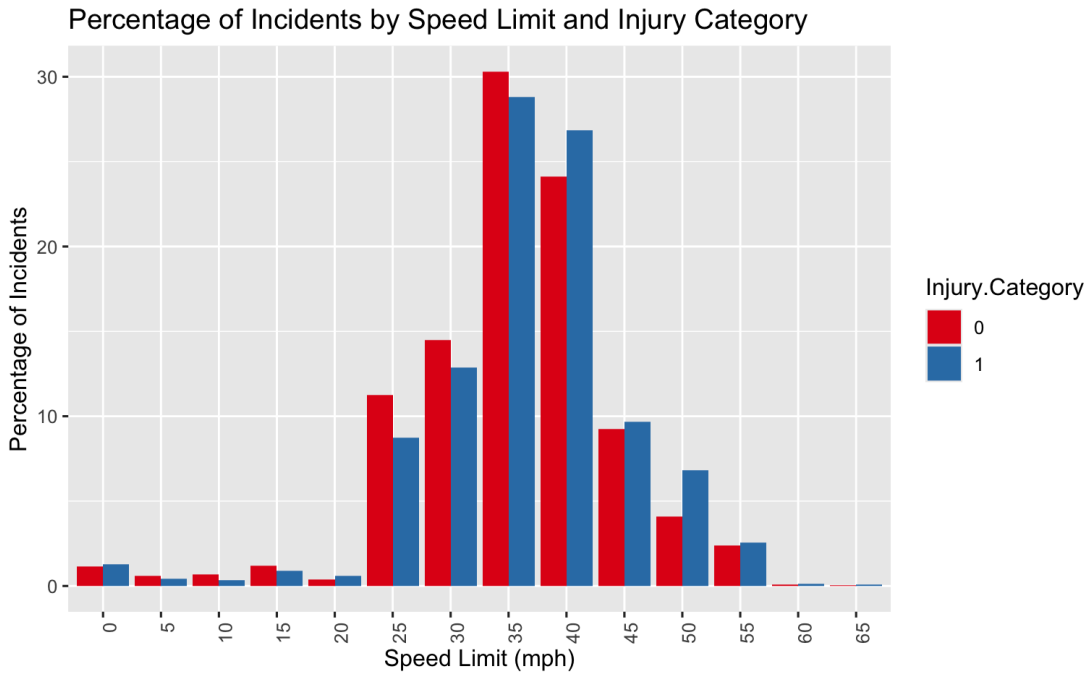

Figure 1. Bar plot of the speed limit to injury category

As the Y variable is binary, we chose to use the bar plot to analyze the relationship between the speed limit and injury severity. Figure 1 visualizes the percentage of incidents by speed limit from 0 mph to 65 mph, and the injury category: suspected minor injury defined as 0, suspected serious injury and fatal injury defined as category 1. From the plot, we find that the majority of incidents happened between the speed limit of 25 mph to 45 mph. Very few incidents happen during the speed limit of 0 mph to 20 mph, and 60 mph to 65 mph. The 0 mph means the car accident happened when the car was parked and crashed by others. The highest percentage of incidents occurs at the speed limit of 35 mph, where the percentage is about 30%. At a moderate speed limit (0-35mph), there is a higher likelihood of incidents involving suspected minor injury compared to suspected serious and fatal injury, and vice versa at a high-speed limit (40-65 mph). Based on the result, we could conclude that there is a significant interaction between the speed limit and injury severity. Lower percentages of incidents at very low (0-20 mph) and very high (45-65 mph) speed limits might suggest that either fewer cars are traveling at these extremes or drivers are more cautious.

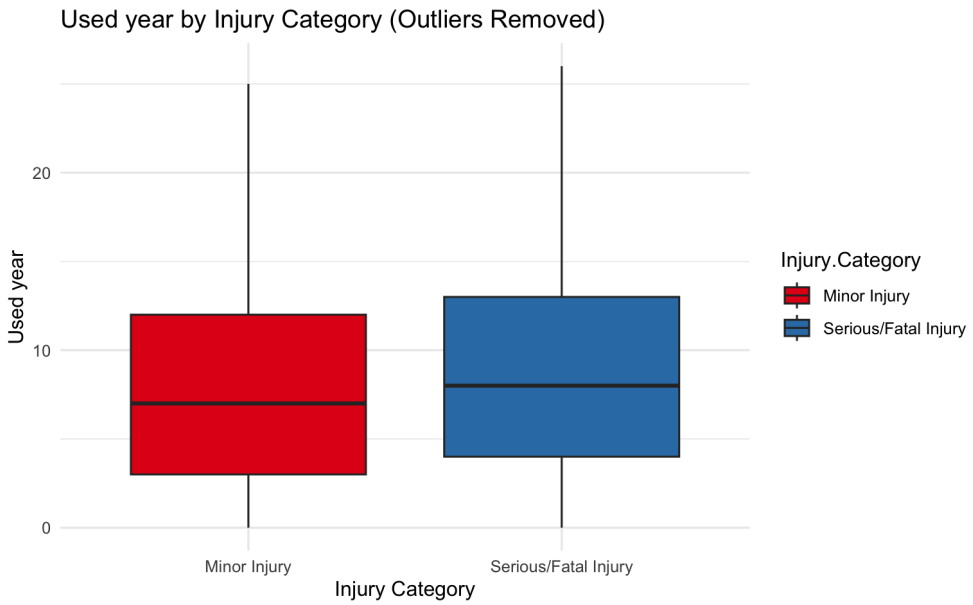

Figure2: Box plot of Used Years to Injury Category

To analyze the relationship between the Used Years (accident occurs year minus the Vehicle's Year) and injury severity, we chose to use the box plot. In this plot, the Minor injury corresponds to the suspected minor injury from our data, and the Serious/Fatal injury corresponds to the suspected serious injury and fatal injury. Since there are several typos in the data, we removed those as outliers and created the box plot to include the variable Used Years from 0 to 26. According to Figure 2, we find that the median of Used Year from Minor Injury indicates a more recent year of vehicle's production than the Serious/Fatal Injury; the higher broader IQR of serious/fatal injuries to minor injuries suggests that there are greater used years linked to serious/fatal injuries. Both categories have vehicles spanning a wide range of used years, however, compared to minor injuries, serious/fatal injuries generally involve older cars. According to the analysis, we could conclude that there is a correlation between the use year of the vehicles and the seriousness of the injuries sustained in collisions: older cars possibly being more likely to cause fatal or seriously injured collisions.

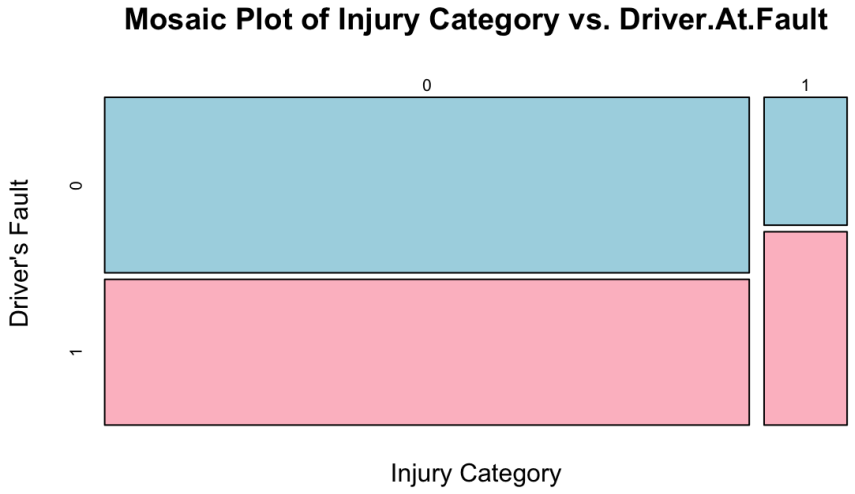

Figure3. Mosaic Plot of Driver's Fault to Injury Category

As both the predicted variable and Y variable are binary, we chose to use the Mosaic plot to analyze the relationship between Driver's Fault and injury severity. In this plot, Injury Category 0 corresponds to the suspected minor injury from our data, and Category 1 corresponds to the suspected serious injury and fatal injury. We defined the Driver's Fault as responsibility for the car accident as 1, and no responsibility as 0. According to Figure 3, in the Injury Category (0), we find that the Driver's Fault is evenly distributed. However, in the Injury Category (1), the Driver's Fault (1) has a larger proportion than the Driver's Fault (0). We could conclude that the Driver's Fault has a relationship to the Injury Category: when the injury severity is suspected serious injury and fatal injury, the Driver's Fault has a larger impact compared to the suspected minor injury.

IV. Methodology

The goal of this study is to display and analyze the graphical relationship between covariate variables and response variables. To achieve this goal, we use R studio as the coding tool to plot and analyze the dataset: Crash Reporting_Drivers Data published on data.gov.
We plan to build a model between our chosen variables from our data. We are choosing one model from Logistic Regression(GLM), Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA). Since the response variable is binary, we choose GLM, where the formula is $p(X) = \frac{e^{\beta 0+\beta 1X}}{1+e^{\beta 0+\beta 1X}}$.

LDA and QDA are two classic classifiers. LDA is more computationally efficient because it requires estimating fewer parameters, and QDA offers more flexibility by modeling non-linear relationships. Both LDA and QDA can be derived from simple probabilistic models which model the class conditional distribution of the data $P(X|y = k)$ for each class k, predictions can then be obtained by using Bayes' rule: $P(y = k|x) = \frac{P(x|y=k)P(y=k)}{P(x)}$. The formula for LDA is: $logP(y = k|x) =- \frac{1}{2}(x - \mu_k)^t\Sigma^{-1}(x - \mu_k) + logP(y = k) + C$, and the formula for QDA is: $logP(y = k|x) = logP(x|y = k) + logP(y = k) + C$.

For the model selection, we use K-fold cross-validation as the first step, which is an approach used to estimate the test error. K-fold cross-validation allows every data point to be used for both training and validation. This method splits the dataset into K equal folds randomly, sets each fold once as validation and sets the remaining folds as the training datasets which are used to fit the model. We repeat this process K times and set a different fold as the validation set each time. We define the K folds be C_1, C_2, … C_k, where C_k denotes the indices of the observations in fold k. In fold k, there are n_k observations. Since our response variable is binary, we choose to analyze the test error rate. The compute formula is $CV_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} Err_k$, where $Err_{test} = \frac{1}{m}\sum_{i=1}^{m} 1_{\{y_i \neq \hat{f}(x_i)\}}$. In our data, the serious and fatal injury was 1482 cases and minor injury was 11509 cases, which is a large sample, so we chose our k to be 50.

We also compare the three models' accuracy from confusion matrices. Since our k is 50, we have 50 confusion matrices for each model. We combine these 50 confusion matrices into one, so we can get one accuracy for each model. Therefore, we will select the model with the highest accuracy.

Then we use Bootstrap, which is a solid statistical methodology that works especially well with large datasets. By using Bootstrap, we can compare the stability of different models. This methodology is invaluable in navigating the intricacies of our data, guaranteeing a thorough comprehension of critical indicators. We aim to confirm the result of model selection in this step.

Finally, we determine the level of correlation between three important factors as part of our research. This understanding can inform the design of our models by highlighting which variables may influence each other. We aim to find out the relationship that may exist between these variables, which may improve the performance of our model.

V. Main results

i) Summary of each model

From the GLM model summary in Table 1, the coefficients of Used. year is 0.02, the Driver.At. fault is 0.60, Speed. Limit is 0.02. The positive coefficient indicates all predictors have a positive relationship with response variables. And all p-values are small indicating that all predictors are statistically significant.

Table 1: Summary of Logistic Regression Model

	Estimate	**Std. Error**	**z value**	**Pr(>|z|)**
(Intercept)	-3.194122	0.132898	-24.034	$< 2e-16$ ***
Used.year	0.020036	0.003943	5.081	3.75e-07 ***
Driver.At.Fault1	0.605925	0.056651	10.696	$< 2e-16$ ***
Speed.Limit	0.018065	0.003245	5.567	2.58e-08 ***

From the summary of the LDA model, we can get the prior probabilities for suspected minor injury is 0.885, and for suspected serious and fatal injury is 0.115. In Table 2, the coefficients of Used. year is 0.06, the Driver.At. fault is 1.66, Speed. Limit is 0.05. The highest coefficient in Driver. At.Fault, indicating it has the most substantial impact on the LDA. All predictors have a positive coefficient, indicating that increases in these variables are associated with the classes defined by LDA. Table 3 shows the means of each group.

Table 2: Coefficients of LDA

Variable	**LD1**
Used.year	0.06169364
Driver.At.Fault1	1.65999227
Speed.Limit	0.04834493

Table 3: Group means

Group	**Used.year**	**Driver.At.Fault1**	**Speed.Limit**
0	8.260426	0.4538598	35.35093
1	9.309166	0.6025992	36.43981

The output of the QDA model also has the mean of each group, prior probabilities of groups, which are the same as the LDA model. However, it does not contain coefficients, because the QDA classifier involves a quadratic, rather than a linear.

ii) K-fold cross-validation

In Table 4, we compare the average error rate and accuracy for GLM, LDA, and QDA models. The result shows that GLM and LDA models have the same average error rate and accuracy. The average error rate of GLM and LDA models is slightly lower than QDA and accuracy is slightly higher than QDA. We conclude that both GLM and LDA are both fittable models. Additionally, we search for the limitations of both models. First, GLM does not assume any specific shapes of densities in the space of predictor variables but LDA does. Second, LDA is useful when n is small, or the classes are well separated, but our data has large datasets and are highly biased. Based on those reasons, our final decision is to choose the GLM model.

Table 4: Average Error rate and Accuracy for different models

result	GLM	LDA	QDA
Average Error Rate	0.1128185	0.1128185	0.1152124
Accuracy	0.8871815	0.8871815	0.8847876

iii) Bootstrap

In the K-fold CV section, we decide to choose the GLM model. For a double-check, we continue to do the Bootstrap for GLM and LDA models. When estimating a statistic's distribution, bootstrap is a flexible technique that works particularly well in cases where the underlying distribution is complex. Because our data is large, we use Bootstrap to provide valuable insights into the reliability and variability of statistical estimates based on our models.

From Bootstrap, the mean accuracy of the GLM model is 0.8859275, and of the LDA model is 0.8859167. It is clear that both models perform effectively. The GLM model has a slightly higher accuracy, which suggests that the GLM model is the best-fitted model for our data.

vi) Correlation

To gain further understanding, we choose to examine the relationship between three variables in order to determine how independent they are of one another. In statistical analysis, correlation serves as a measure of the statistical association between two random variables, essentially indicating their degree of dependence.

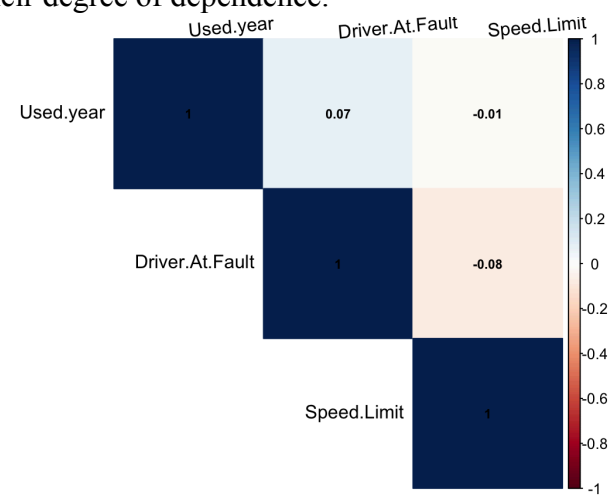

Figure4: Correlation Plot of Predictor Variables

As can be seen from the analysis in Figure 4, the absolute value of correlation coefficients between every two variables are all smaller than 0.1, which indicates that there are weak correlations between these three variables. This supports the idea of their relative independence by implying that they don't have significant linear ties with one another.

Hence our final model is: $p(X) = \frac{e^{-3.194122+0.020036X_{used.year}+0.605925X_{driver.fault}+0.018065X_{speed.limit}}}{1+e^{-3.194122+0.020036X_{used.year}+0.605925X_{driver.fault}+0.018065X_{speed.limit}}}$.

VI. Discussion

The analysis shows all the used years of vehicle, driver's fault, and speed limit are positively related to fatal and serious injuries. There are three limitations of our approach that we need to improve in our future research, (1) the data is not well-structured, (2) shortage of predictors, (3) only includes multivariate. Typo of recording and outliers cause poorly structured, which might lead to a biased result and influence the statistical model. The number of predictors,

which is 3 out of a total of 43 variables including the response variable, is limited. The limited number of predictors may not capture the full complexity of factors that influence the risk of serious and fatal injuries. Hence the model may not have a nice predicted accuracy. Overlooking "Risk and Protection Factors in Fatal Accidents" from Dupont et al (2010) and "Etiology of Fatal THORACIC AORTIC INJURIES: Secondary Data Analysis" from Sznol et al (2015), two researchers both use univariate analysis and multivariate analysis, only focusing on the multivariate analysis may lead to missing important insights into each variable's behavior and distribution. In addition, one thing that is unique in this study is that we choose the k for k-fold CV is 50, but commonly people use 5 or 10 for the k.

On the other hand, the shortage of predictors could be seen as focusing on the specific factors that affect the risk of serious and fatal injuries, which could become one strength of our approach. Besides focusing on specific factors, our strength is the comparison of different models. Different from "Risk and Protection Factors in Fatal Accidents" published by Dupont et al (2010) which only use binomial logistic regression, we compare GLM, LDA, and QDA by coding to choose the fittest model. This method helps us to figure out the best model as well.

In the future, we need to improve our approach and go deep on this topic. We plan to search the current law to see if there are any measures being taken. Also, we will analyze other state's data to do the comparison to see the influence of different laws. Additionally, we will check the difference in injuries for different accident years. For example, we could analyze if Covid-19 affects the amounts of injuries.

VII. Conclusion

In conclusion, this study aims to analyze the relationship between the predictors of speed limit, vehicle year, driver's fault, and the response variable, injury severity. By defining the vehicle's used year as the difference between the accident occurred year and the vehicle's production year, we gained insights into the impact of vehicle age on injury severity.
We first create three different plots based on each predictor as visualizations to see the data distribution. Visualizations demonstrated that the moderate speed limit is associated with an increased rate of minor injury incidents, while severe injuries are correlated with high-speed limits. Serious or fatal collisions tend to be associated with older vehicles, and driver fault is a significant determinant of injury severity.

Then we use k-fold cross-validation to estimate three models: GLM, LDA, and QDA. By comparing the average error rate and accuracy, we found that GLM and LDA have the same average error rate and accuracy, which are slightly better than QDA. However, due to the dataset's size, and bias, we determined that the GLM model is the most suitable for this analysis. And by comparing accuracy through bootstrap tests, the conclusion based on K-fold CV has been confirmed. We also examine the correlation between three predictor variables, and results show these three variables are independent. Ultimately, the study confirms the positive relationship between the predictors (speed limit, vehicle year, and driver's fault) and injury severity, highlighting their significance in understanding and mitigating road traffic injuries.

In conclusion, our study provides valuable insight into the factors influencing injury severity in car crashes. By understanding these factors, we can better design interventions to improve road safety and reduce the severity of injuries, ultimately saving lives and reducing the burden on the healthcare system.

Reference

Chen, K. L., Tsai, B.-W., Fortin, G., & Cooper, J. F. (2018, September 17). SafeTREC Traffic Safety Facts: Speeding-Related Collisions. Berkeley SafeTREC. https://www.ots.ca.gov/wp-content/uploads/sites/67/2019/06/2018-Speeding-Related-Collisions-Facts.pdf

Driver error accident, Florida lawyer. Law Offices of Brent C. Miller, P.A. (2021, October 8). https://www.bcmillerlaw.com/auto-accidents/driver-error/#:~:text=Examples%20of%20driver%20error%20may,falling%20asleep%20at%20the%20wheel

Dupont, E., Martensen, H., Papadimitriou, E., & Yannis, G. (2010). Risk and protection factors in fatal accidents. *Accident Analysis & Prevention*, *42*(2), 645–653. https://doi.org/10.1016/j.aap.2009.10.011

Fatality facts 2021: Yearly snapshot. IIHS. (2023, May). https://www.iihs.org/topics/fatality-statistics/detail/yearly-snapshot

Historical car crash deaths and rates. Injury Facts. (2024, April 25). https://injuryfacts.nsc.org/motor-vehicle/historical-fatality-trends/deaths-and-rates/

Høye, A. (2019, February). Vehicle registration year, age, and weight – untangling the effects on crash risk. Accident Analysis & Prevention, 123, 1–11. https://doi.org/10.1016/j.aap.2018.11.002

Karlin, M. (2024, May 9). *California car accident statistics 2024 - traffic fatalities Ca*. Karlin & Karlin, APC. https://www.karlaw.com/blog/california-car-accident-statistics/#:~:text=At%20least%20250%2C000%20accidents%20in,pedestrians%20(around%201%2C000%20annually).

Publisher data.montgomerycountymd.gov. (2024, February 9). *Montgomery County of maryland - crash reporting - drivers data*. Catalog. https://catalog.data.gov/dataset/crash-reporting-drivers-data

Sznol, J. A., Koru-Sengul, T., Graygo, J., Murakhovsky, D., Bahouth, G., & Schulman, C. I. (2015). Etiology of fatal THORACIC AORTIC INJURIES: Secondary Data Analysis. *Traffic Injury Prevention*, *17*(2), 209–216. https://doi.org/10.1080/15389588.2015.1067805